

Square Roots & Perfect Squares

Grade 4 Mathematics Lesson

Learning Objectives

- Understand what a **square** is in math.
- Learn about **perfect squares**.
- Discover what **square roots** are.
- Practice finding square roots in real-life examples.

What is a Square?

In math, "squaring" a number means multiplying that number by itself!

Example: $2 \times 2 = 4$

When we multiply 2 by itself, we get 4. This is called squaring the number 2.

Visualizing Squares

Think of dots arranged in a perfect square pattern:

$1 \times 1 = 1$ (One tiny dot square)

$2 \times 2 = 4$ (2 rows of 2 dots)

$3 \times 3 = 9$ (3 rows of 3 dots)

Perfect Squares

A **perfect square** is the result you get when you multiply a whole number by itself.

The First Ten Perfect Squares:

1, 4, 9, 16, 25, 36, 49, 64, 81, 100

How they are made:

- $1^2 = 1$

- $2^2 = 4$

- $3^2 = 9$

- $4^2 = 16$

- $5^2 = 25$

- $6^2 = 36$

What is a Square Root?

Finding a square root is like **going backwards!** It asks: "What number multiplied by itself gives me this result?"

The symbol for square root is $\sqrt{}$.

Example: $\sqrt{9} = 3$

We say: "The square root of 9 is 3" because $3 \times 3 = 9$.

Practice Reading Symbols:

- $\sqrt{16}$ means "Square root of 16" (Answer: 4)

- $\sqrt{25}$ means "Square root of 25" (Answer: 5)

Squares in Real Life

You can find perfect squares all around you!

Floor Tiles: Many kitchen or bathroom tiles are perfect squares arranged in grids.

Chessboards: A chessboard is an 8×8 grid, making a total of 64 small squares!

Window Panes: Look for square glass panes arranged in rows and columns.

Quick Quiz & Practice

1. **What is 5 squared?** (Hint: Multiply 5 by itself!)

Answer: _____

2. **What is $\sqrt{16}$?** (Hint: Which number times itself is 16?)

Answer: _____

3. **Bonus: Is 20 a perfect square?**

Answer: _____

Key Takeaways

- **Squaring:** Multiplying a number by itself ($4 \times 4 = 16$).
- **Perfect Squares:** The results (1, 4, 9, 16, 25...).
- **Square Roots:** Finding the original number ($\sqrt{16} = 4$).

Math is Amazing! You did great!